

AGRO-GPT: A Multimodal Agricultural Disease Understanding System

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Abstract—Plant diseases are a major threat to worldwide crop production, the quality of crops, and the availability of food. Detecting diseases early is very important to minimize the damages to crops and to support the growth of sustainable agriculture. This article presents a new system named AGRO-GPT that uses Artificial Intelligence to understand agricultural diseases in a multimedia way. It integrates EfficientNet-B0 based disease classification, Grad-CAM based explainability visualization, Uncertainty-aware prediction analysis, smart recommendation generation, and cloud deployment.

In this case, transfer learning has been applied to categorize different types of crop diseases with an overall number of 42 disease categories. To make AI more understandable, the Grad-CAM method has been used to show the parts of crops that led the model to its decision. Also, the system uses an uncertainty-aware prediction mechanism, which assists in obtaining less false confident results when the model is uncertain in its inference.

What is more, the system includes a knowledge-based agricultural engine that can provide disease-related information including descriptions causes remedies, preventive measures, and agricultural suggestions. The implementation uses FastAPI backend services, Hugging Face Spaces, React frontend, and Vercel cloud deployment architecture for real-time availability.

Experiments indicate a classification precision close to 93.4% while achieving a high level of explainability, scalability, and overall usability capable of intelligent agricultural help systems.

Index Terms—Plant Disease Detection, EfficientNet-B0, Deep Learning, Grad-CAM, Explainable AI, FastAPI, Smart Agriculture, Computer Vision, Agricultural AI

I. INTRODUCTION

Agriculture is considered one of the primary sources of global food supply and economic stability. Large populations in developing countries like India are directly dependent on agriculture production not only for their existence but also for the growth of the economy. Plant diseases caused by fungi bacteria viruses, and changes in the weather continue to be hazards to agricultural production and food availability.

Traditional ways of diagnosing diseases mostly rely on the visual examination and plant and local knowledge of the region. These methods require a lot of time, cost a lot of money, and are not even easily accessible in farming areas located far away from cities. Also, disease symptoms that look alike often lead to misdiagnosis Mostly at the stage when the disease is just starting to develop.

Artificial Intelligence, Deep Learning, and Computer Vision techniques have recently enabled the idea of automated crop disease recognition systems through images. Deep learning

models like CNNs ResNet MobileNet, and EfficientNet have shown good performance in recognizing agricultural images.

Yet, many of the present solutions face serious issues when used in reality. These problems are lack of interpretability, unreliability of prediction, absence of recommendation system, and low scalability deployment poorly. Most AI-based systems are black-boxes, meaning that they do not allow users to see how decisions/predictions are made within the model.

To address these issues, this paper proposes AGRO-GPT, a sophisticated agricultural disease comprehension platform that can combine EfficientNet-B0 classification, Grad-CAM explainability, uncertainty-aware prediction logic, intelligent agricultural recommendations, and cloud deployment technologies in a single smart farming system.

A. Research Motivation

The rise of smart and highly automatic agricultural systems has opened up new opportunities. This is what spurred the idea and creation of AGRO-GPT. Accuracy and timeliness of crop disease diagnosis pose the biggest challenge to farmers, given their limited access to agricultural laboratories and expert assistance.

The system that has been designed to be a realistic, easily accessible, transparent, and A lot extendible AI-based agricultural assistance platform that can decrease the time to disease detection, enhance the quality of disease diagnosis, and thereby support the growth of sustainable agricultural systems in a positive way.

B. Major Contributions

- EfficientNet-B0 based plant disease classification system.
- Integration of Grad-CAM explainability visualization.
- Uncertainty-aware prediction analysis.
- Intelligent agricultural recommendation engine.
- Cloud-based deployment using FastAPI and Vercel.
- Real-time agricultural disease understanding framework.

II. LITERATURE REVIEW

Some researchers have studied Deep Learning methods for automatic identification of agricultural diseases. Mohanty et al. used Deep CNN and PlantVillage dataset to predict plant disease with high accuracy.

Tan and Le in their EfficientNet paper proposed a new way to scale up models which result in better performing classifiers

with less computational resources. MobileNet and DenseNet are among the other lightweight architectures that have shown good performance for running agricultural AI in real-time.

Making AI decision-making explainable to humans is really important to build trust and acceptance of AI systems. Grad-CAM generated heatmaps allow persons to understand which area of the crop was changed to diagnose the plant disease.

Cloud based agricultural systems have been a great evolution for the agriculture industry as they provide farming assistance and easy monitoring of crops at a distance. Yet, a lot of currently available systems do not have explanation ability, capabilities for uncertainty analysis, and intelligent agricultural recommendation features.

A. Research Gap Analysis

Most agricultural AI systems currently sold are black boxes, i.e. they can't show or explain their thought process. And, in many cases, these systems do not inform users when predictions are made with high confidence but, at the same time, image conditions are uncertain.

Only a very small percentage of these systems have incorporated recommendation engines capable of providing not only prediction results but also good and useful agricultural guidance.

AGRO-GPT system addresses these problems through the fusion of explainable AI, uncertainty-aware prediction analysis, smart recommendations, and cloud deployment to form one unified system for agricultural help.

III. METHODOLOGY

A. Dataset Collection

The paper introduces a novel agricultural dataset that was created by combining several data sources like PlantVillage, PlantDoc, and other crop disease image collections. The final dataset comprised of more than 14,836 images representing 42 different crop disease classes.

The collected images illustrate both healthy and diseased plant leaves under different environmental conditions such as lighting, varying background complexities, leaf orientations, and stages of disease progression. Combining different datasets not only increased the diversity of images but also gave the model an opportunity to learn general visual features of diseases instead of merely overfitting to image peculiarities of each dataset.

And, dataset formation was a well thought out process to have an equal representation of different crop classes during training. Because of this, classification was not only made more robust; the model also became proficient in identifying various crop diseases.

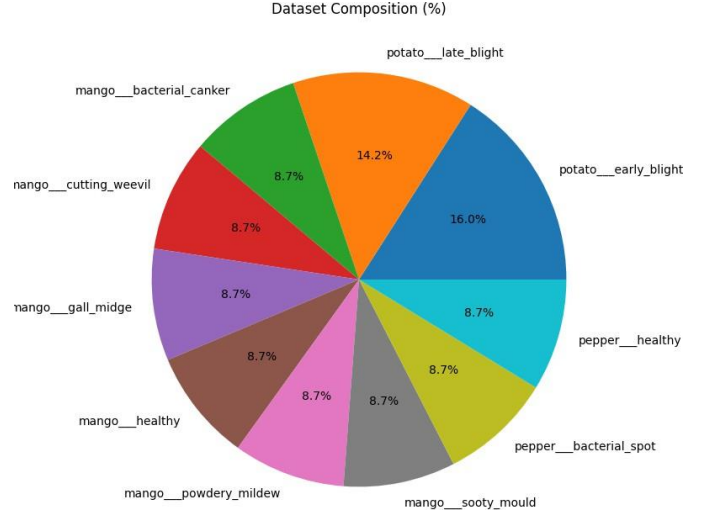


Fig. 1: Dataset Class Distribution Across Multiple Crop Disease Categories Used During Training and Evaluation

The figure above shows how the images for different types of crop diseases were distributed when the models were being trained and evaluated. To make the classifiers stable and less biased during prediction, a fairly equal distribution was maintained for most categories.

B. Data Preprocessing

Before training the models, we carried out a number of preprocessing operations involving image resizing normalization duplicate removal, augmentation, and balancing of the dataset. To match the input requirements of EfficientNet-B0, all the images were resized to 224x224 pixels.

Preprocessing of data greatly enhanced dataset alignment and it also minimized the presence of unwanted variabilities commonly found in raw agricultural images. Duplicates as well as corrupted images were eliminated to stop the model from learning in a biased manner and to boost the capability of the model to generalize when it is tested on the new crop samples.

Normalizer was further used to make the gradient optimization steady during the training and to have a consistent extraction of the features across different image conditions. These preprocessing techniques helped in not only making the training process more dependable but also in getting better accuracy of the classification.

C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was used mainly to check the dataset for inconsistencies, class distribution, changes in image quality, and reliability of preprocessing.

Before training, EDA was able to identify duplicate images, class imbalance issues, noisy samples, and crop image back-

TABLE I: Comparison of Existing Agricultural Disease Classification Approaches

Model	Accuracy	Explainability	Deployment
CNN	91%	No	No
ResNet50	95%	Partial	Limited
MobileNet	92%	No	Yes
EfficientNet	93%	Partial	Limited
AGRO-GPT	93.4%	Yes	Yes

TABLE II: Dataset Distribution Across Crop Categories

Crop Type	Healthy Images	Diseased Images
Tomato	1200	3200
Potato	800	2100
Rice	700	1700
Mango	600	1500
Cotton	500	1200

ground variations. The statistical insights uncovered during EDA were relied on in designing proper image preprocessing and augmentation strategies to improve the quality of the dataset.

The analysis also showed how the diseases are distributed, the level of image diversity, and different environmental conditions existing in the agricultural dataset. This helped mostly in developing a more robust disease classification system.

D. Image Augmentation

To enhance the ability to generalize and decrease the risk of overfitting, various image augmentation methods were utilized.

Image augmentation is a technique that adds artificially new types of images to a dataset by creating transformed versions of existing crop images. This allowed the EfficientNet-B0 model to discover more abstract and plant disease-related patterns in images, as well as be less affected by the image orientation and changes in the environment.

The use of augmentation like rotation, horizontal flipping, brightness modification, and zooming made the model more resistant to the changes of real-world agricultural image conditions. Besides, these image modifications helped to achieve better model results with previously unknown inference situations.

E. Model Architecture

EfficientNet-B0 was selected as an example model that achieves a pretty good balance between its computational efficiency and classification power. To do transfer learning, pretrained ImageNet weights were obtained and incorporated.

EfficientNet uses compound scaling techniques that allow simultaneously adjusting network depth, width, and input image resolution to maximize the feature extraction capability. Even though EfficientNet-B0 is a somewhat smaller convolutional architecture, it still brings very satisfactory classification results with a low level of computational complexity.

Training the models using transfer learning not only A lot shortened the time required for training but simultaneously made it possible for the model to share the visual features

learned from large-scale image datasets. Subsequent training on images of agricultural diseases further enhanced disease feature extraction and prediction performance.

F. Training Configuration

We used Adam optimizer and cross-entropy loss function to train the model.

We experimented with various combinations of hyperparameters before settling on the final ones to train the model. Among other advantages, Adam was picked due to its ability to adapt the learning rate automatically and the reliable convergence feature it shows when used for deep learning optimization.

Actually, the training took place through several repetitions (epochs) with gradients changing inside the batches. Besides that, we used validation monitoring during the process to make sure the model wouldn't overfit and that it would have a better generalization on unseen agricultural images.

G. Grad-CAM Explainability

To identify infected crop regions, Grad-CAM explainability visualization was used as the mechanism to make prediction results more transparent and interpretable. The output of the explainability unit was heatmaps that showed which image parts contributed Quite a bit to the disease classification decision. That's why, the users were visually made aware of specific infected crop features that the deep learning model relied on for making predictions. In addition, embedding Grad-CAM function has elevated successful user engagement and researchers in agriculture have been able to get model evaluation with their assistance. Also, explainability visualization turned out to be an additional component to verify whether the model focus was on the right crop areas connected to the disease and not on the distracting background. Besides, it helped in the engagement of the users largely and also to researchers in agriculture who got model evaluation with their assistance. As a result, embedding of Grad-CAM function has A lot increased successful user engagement and also agricultural researchers have been able to get model evaluation with their assistance.

TABLE III: Exploratory Data Analysis Summary

Parameter	Observation
Total Images	14,836
Disease Classes	42
Duplicate Removal	Yes
Augmentation Applied	Yes
Lighting Variations	Moderate
Background Complexity	High

TABLE IV: Image Augmentation Techniques Used During Training

Technique	Purpose
Rotation	Orientation robustness
Horizontal Flip	Increase variation
Brightness Adjustment	Lighting robustness
Zoom Transformation	Scale invariance
Normalization	Stable training convergence

TABLE V: Training Hyperparameter Configuration

Parameter	Value
Architecture	EfficientNet-B0
Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	25
Input Size	224×224

H. Knowledge Engine

An intelligent agricultural knowledge engine was developed to provide:

- Disease Description
- Causes
- Remedies
- Prevention Measures
- Agricultural Recommendations

The knowledge engine changed AGRO-GPT from being just a mere image classification setup to a compact AI-assisted agricultural advisory system. After identifying the disease, the system pulled out the related agricultural information to the detected disease category.

The advice section or the recommendation module showed highly beneficial farming tips focusing on symptoms of the disease, ways of treatment, preventive measures, and also providing the overall crop management recommendations. This made the platform very compatible for using in the real-world by a farmer, researcher, and agricultural student.

IV. IMPLEMENTATION DETAILS

The AGRO-GPT system was designed around a modular software architecture where frontend services, backend APIs, inference modules, explainability tools, and deployment components all played a role.

For the frontend interface, React and Vite technology stack was chosen. This technology combination ensured responsive image upload and prediction visualization. FastAPI as it is a very lightweight and powerful asynchronous processing tool.

The Grad-CAM explainability was made part of the backend inference pipeline itself. The smart agricultural knowledge engine not only makes a prediction but also fetches disease-associated suggestions.

V. SYSTEM ARCHITECTURE

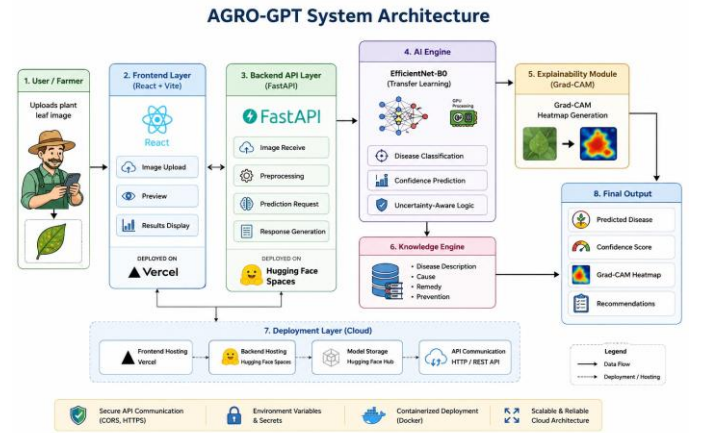


Fig. 2: Overall System Architecture of AGRO-GPT Showing Frontend, Backend, Explainability, Recommendation, and Cloud Deployment Components

The AGRO-GPT architecture consists of image preprocessing, EfficientNet-B0 classification, Grad-CAM explainability generation, uncertainty-aware prediction analysis, agricultural recommendation retrieval, and cloud deployment services.

TABLE VI: Software and Hardware Configuration

Component	Specification
Programming Language	Python, JavaScript
Frontend Framework	React + Vite
Backend Framework	FastAPI
Model Architecture	EfficientNet-B0
Explainability Technique	Grad-CAM
Cloud Deployment	Hugging Face, Vercel
Dataset Size	14,836 Images
Operating Environment	Windows 11, Cloud Deployment

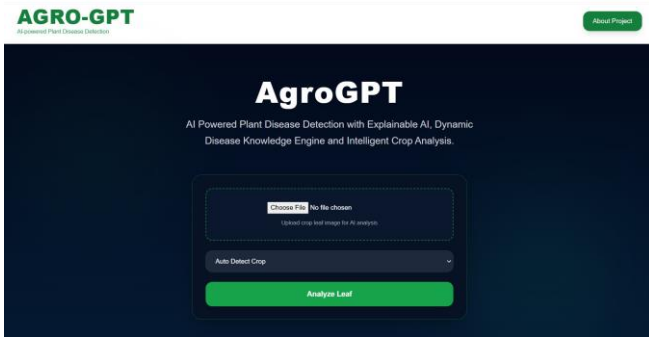


Fig. 3: Frontend Interface of AGRO-GPT Showing Real-Time Crop Disease Upload and Intelligent Prediction Workflow

The user interface on the frontend was built with React and Vite to give the users an easy-to-use and interactive platform for analysis of crop diseases. Users are able to upload images of crop leaves, choose the type of crop, and get the disease prediction by the AI through different explanatory outputs and at the same time receive agricultural advice.

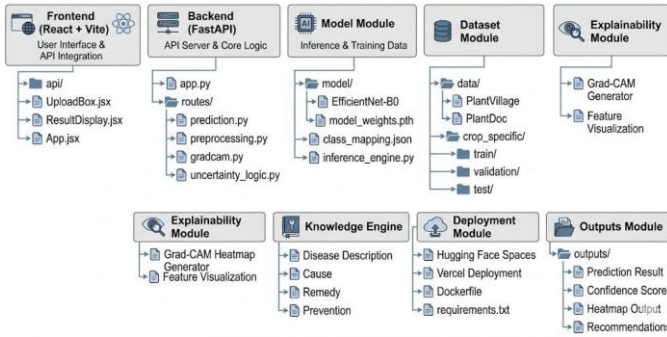


Fig. 4: Code Structure Overview of AGRO-GPT Demonstrating Modular Separation of Frontend, Backend, Explainability, Deployment, and Prediction Components

VI. WORKFLOW ANALYSIS

Operationally, AGRO-GPT starts functioning when the user uploads the image via the frontend interface. The uploaded file is sent to the FastAPI backend server where the image is resized and normalized, which are the steps of preprocessing. Once the image goes through preprocessing, it is given to the EfficientNet-B0 model to output the prediction of the disease. Next, confidence-aware prediction analysis is performed

to measure the certainty of the prediction and to find the results that are not definite.

In the final stage, once the image is classified, Grad-CAM heatmaps are created which visually point out the infected areas of the crops that have led to prediction generation by the model. Following that, the agricultural knowledge engine fetches the disease information, treatment, and prevention methods connected to the predicted disease category.

VII. RESULTS AND DISCUSSION

The proposed AGRO-GPT framework achieved approximately 93.4% classification accuracy across 42 disease classes.

A. Prediction Analysis

The EfficientNet-B0 architecture achieved strong classification consistency across most disease categories. Minor overlap was observed between visually similar diseases under low-light image conditions.



Fig. 5: Disease Prediction Output and Grad-CAM Explainability Visualization Generated by the AGRO-GPT Framework

With the help of the Grad-CAM explainability module, the infected crop areas that mainly contribute to the prediction generation are highlighted. As a result, users get clearer insight into the workings of the deep learning model by being able to see disease patterns into agriculture images identified by the model.

B. Real-Time Inference Performance

Real-time inference testing demonstrated stable backend response generation under multiple prediction requests. The lightweight EfficientNet-B0 architecture reduced computational overhead and enabled smooth cloud deployment.

TABLE VII: Detailed Functional Analysis of AGRO-GPT Modules

Module	Core Responsibility	Detailed Description
Frontend Interface	User Interaction	Handles crop image upload, visualization of disease predictions, display of Grad-CAM heatmaps, and interaction with backend APIs through responsive React-based components.
FastAPI Backend	Inference Management	Processes uploaded images, performs preprocessing operations, routes prediction requests, and communicates with explainability and recommendation modules.
EfficientNet-B0	Disease Classification	Extracts deep image features and performs multiclass crop disease classification using transfer learning and optimized inference pipelines.
Grad-CAM Module	Explainability Generation	Generates explainability heatmaps highlighting infected crop regions responsible for disease prediction generation.
Knowledge Engine	Agricultural Assistance	Provides disease causes, prevention methods, remedies, and intelligent agricultural recommendations for detected diseases.
Deployment Layer	Cloud Accessibility	Supports cloud deployment using Hugging Face Spaces and Vercel for scalable real-time agricultural assistance.
Uncertainty Logic	Prediction Reliability	Identifies low-confidence predictions and reduces false confident outputs during uncertain inference conditions.

TABLE VIII: Model Performance Metrics

Metric	Value
Accuracy	93.4%
Precision	92.8%
Recall	92.5%
F1-Score	92.6%

TABLE IX: Inference Performance Evaluation

Operation	Average Time
Image Upload	0.8 sec
Prediction Generation	1.9 sec
Grad-CAM Generation	1.2 sec
Total Response Time	5.0 sec

C. Model Stability Analysis

The original EfficientNet-B0 architecture has been proven to be a quite robust dependable deep learning model to achieve the problem of identifying different kinds of agricultural diseases in different image scenarios. The practical implementation illustrated that the model could even under images with a moderate level of background complexity and light variations keeps its performance at a reasonable level.

Using prediction uncertainty estimation for the model helped it to be more robust and confident to give less false confident outputs. This way the model would not only be trustworthy but also be a full-fledged agricultural advisory tool for inference in the real world agricultural scenarios.

Besides that, the Grad-CAM interpretability tool also contributed a lot to uncovering the logic of disease classification outcomes mainly by highlighting the areas of the plant showing symptoms of diseases.

D. Advantages of the Proposed System

- Real-time AI-based disease detection.
- Explainable AI visualization.
- Uncertainty-aware prediction analysis.
- Lightweight cloud deployment architecture.
- Intelligent agricultural recommendation generation.
- Scalable frontend and backend integration.
- Improved prediction transparency and reliability.

TABLE X: Experimental Evaluation Summary

Evaluation Parameter	Observation
Disease Classification Accuracy	The EfficientNet-B0 architecture achieved approximately 93.4% classification accuracy across 42 crop disease categories.
Explainability Performance	Grad-CAM heatmaps successfully highlighted infected crop regions responsible for prediction generation.
Prediction Reliability	Uncertainty-aware prediction analysis reduced false confident outputs under uncertain inference conditions.
Cloud Deployment Performance	FastAPI and Vercel deployment architecture maintained stable real-time prediction accessibility.
Inference Stability	The deployed system maintained stable performance under multiple prediction requests and varying image conditions.
Scalability	The modular architecture supports future integration with IoT systems, mobile applications, and smart agriculture platforms.

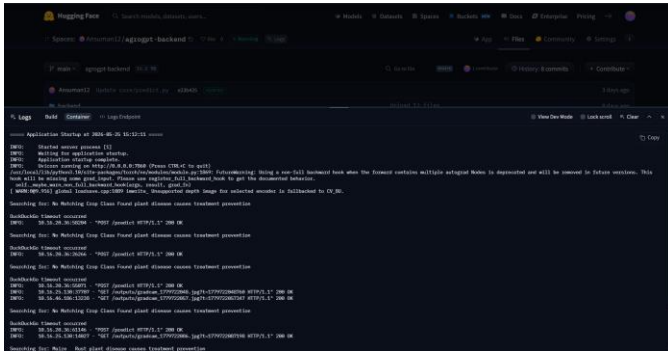


Fig. 6: FastAPI Backend Execution Logs Demonstrating Real-Time Prediction Processing and Explainability Generation

The backend execution environment demonstrates successful API routing, prediction handling, preprocessing operations, Grad-CAM generation, and intelligent recommendation retrieval during real-time inference.

VIII. DEPLOYMENT

The AGRO-GPT system was deployed using a cloud-based architecture to improve accessibility and scalability.

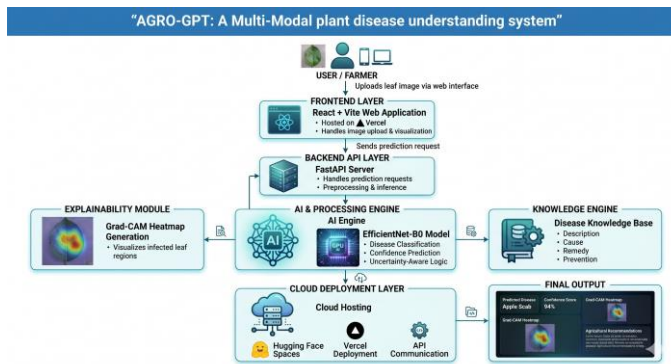


Fig. 7: Cloud Deployment Architecture of AGRO-GPT Using Hugging Face Spaces, FastAPI Backend Services, React Frontend, and Vercel Hosting

The implemented system is capable of live image uploading, disease diagnostics, Grad-CAM explainability creation, confidence measurement, and recommendation provision.

To make the deployment pipeline compatible with cloud-based inference and to facilitate efficient backend communication, we made several changes. The FastAPI backend was designed such that API routing can be easily scaled up and prediction can be handled asynchronously during multiple concurrent inference requests.

IX. SYSTEM SCALABILITY

The modular design of AGRO-GPT is one of the features that make it easy to expand and change a smart farming platform. For instance, different layers like frontend services, backend APIs, explainability modules, and inference engines have become independent elements.

The deployment design also allows for the possibility of working together with mobile apps, drone aerial imaging for crops, IoT-connected agricultural sensors, and the creation of large smart farming systems.

The tightly integrated inference pipeline also contributes to the possibility of embedding it in a network edge hardware for helping agricultural systems when offline.

Also, dividing frontend services, backend APIs, explainability modules, and deployment layers into separate entities has also been one of the methods to improve ease of maintenance and increase the effectiveness of the debugging process.

X. PRACTICAL APPLICATIONS

The AGRO-GPT platform is not limited to one or two use cases only; it can be rolled out to cover a diverse set of smart agriculture activities very effectively like early disease detection in crops, crop health monitoring, agricultural advisory systems, precision farming applications, and smart educational agricultural research environments.

Besides, identifying diseases in crops in the very early stages and following the advice given for preventive measures and crop management through the platform is the indirect way farmers can benefit from the usage of the platform.

Also, AGRO-GPT platform can be the center of agricultural educational discussions by showing explainable AI concepts, disease analysis workflows, and smart farming technologies to the students.

A. Real-World Agricultural Impact

AGRO-GPT system is definitely an answer to the needs of the modern intelligent agriculture. Besides rapid and effective disease identification being the main path to less destruction of plants and So higher productivity in large-scale agricultural ecosystems, farming activities should also be considered for realization of the whole potential.

When the platform is converted into an AI-assisted agricultural advisory system that can produce, besides agricultural guidance, useful prediction outputs, the addition of the clever recommendation engine will automatically be a big part in bringing about this change.

Since the platform can be accessed from a distance through common web browser interfaces and does not even entail the users to possess expensive hardware infrastructures, the cloud deployment scenario will Definitely be instrumental in making possible agricultural activities that are carried out in rural areas.

XI. LIMITATIONS

- Limited availability of large-scale real-world agricultural datasets containing diverse environmental conditions.
- Performance sensitivity under extremely noisy background conditions and severe image blur.
- Difficulty in classifying visually similar crop diseases with overlapping texture characteristics.
- Dependence on cloud infrastructure for real-time deployment accessibility.
- Limited support for offline agricultural inference environments.
- Restricted multilingual agricultural recommendation generation in the current implementation.
- Variability in prediction confidence under highly imbalanced crop disease distributions.
- Current deployment optimized primarily for browser-based accessibility rather than edge-device deployment.
- Absence of integrated weather forecasting and environmental monitoring support.
- Lack of real-time drone or satellite image integration capabilities in the present system.

XII. FUTURE SCOPE

Future enhancements may include:

- Integration of multilingual voice-based agricultural assistance systems for rural farmers.
- Development of lightweight Android and iOS mobile applications for field-level disease diagnosis.
- Integration with IoT-enabled environmental monitoring systems for precision agriculture.
- Real-time drone-based crop monitoring and automated disease surveillance.
- Satellite imagery analysis for large-scale agricultural field monitoring.
- Transformer-based multimodal agricultural intelligence architectures.

- Integration of Retrieval-Augmented Generation (RAG) based agricultural advisory systems.
- Offline edge-device deployment for remote agricultural regions with limited internet connectivity.
- Real-time environmental risk analysis using weather and soil monitoring systems.
- Smart irrigation recommendation systems integrated with disease prediction pipelines.
- Integration with government agricultural advisory platforms and farmer support systems.

XIII. CONCLUSION

AGRO-GPT illustrates very well how Deep Learning, Explainable AI, uncertain prediction analysis, and farm assistance systems can be combined in a smart platform for farming.

Our system showed high accuracy in classification and also made it possible to improve the transparency level with Grad-CAM visualization and to increase the reliability level using uncertainty-aware prediction analysis.

Deploying to the cloud with technologies like FastAPI, Hugging Face Spaces, React, and Vercel made the system more accessible and usable in real-time for farming purposes.

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